

भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 360: Machine Learning Lab
Practice Assignment 7

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. Consider the **Mall Customer dataset** to find the target customer groups for market strategy planning.
   Write a program to do the followings:
   (a) Read the dataset.
   (b) Cluster the dataset into different categories of customers using DBSCAN Clustering method(**without using in-built Python library/ packages**).
   Perform hyper-parameter tuning (using grid-search method) for the following hyper parameters considering the given values:
   - minpts: {4 - 10}
   - epsilon: {2, 2.5, 3, 3.5, 4}

   (c) Determine the optimal value of epsilon using **k-distance graph**, w.r.t. the optimal minpts-value from (b), and then apply DBSCAN clustering. Consider the optimal minpts-value as same as obtained in 1(b).
   Hint: Refer for K-distance graph.
   (d) Report and visualize the obtained final clusters for both 1(b) and 1(c), and write the result analysis.

2. (a) Cluster the above **Mall customer dataset** into different categories of customers using Hierarchical clustering method, with single-linkage, complete-linkage and centroid-linkage strategies separately(**without using in-built Python library/ packages**).
   (b) Report and visualize the obtained final clusters (dendrogram) for all the three strategies, and write the result analysis.